JC2001: Introduction to Software Engineering

Group Project Description and Submission Requirements

Academic Year: 2026-27

The group project is worth 50% of the overall course mark. Table 1 shows the different components and activities of the group project, including those assessed for marks and those that are mandatory but carry no marks.

Table 1: Group Project Assessment Components and Mandatory Requirements

Sr No	Component	Weighting	Requirement
1	Project Proposal	0%	Submission is mandatory by the deadline. A proposal submitted after the deadline but within 7 days will be subject to the applicable late-submission penalty. A proposal submitted more than 7 days after the deadline will be treated as not submitted. Non-submission or late submission may result in a maximum penalty of 10% of the overall group project mark.
2	Technical Report	50% of the group project mark	Assessed submission.
3	Proof-of-Concept Software	30% of the group project mark	Assessed submission.
4	Project Presentation	20% of the group project mark	Assessed submission.
5	Project Updates	0%	Four project updates will be scheduled, and each group must submit at least three. A required update submitted after its deadline but within 7 days will be subject to the applicable late-submission penalty. An update submitted more than 7 days after its deadline will be treated as not submitted. Failure to submit at least three project updates may result in a maximum penalty of 10% of the overall group project mark.

Note: *All submissions are group submissions, so one team member from each group must submit on behalf of their group.*

1. Project Proposal (Mandatory – Unmarked)

Each group must submit a project proposal by the deadline specified in the Course Assessment folder on the MyAberdeen (Blackboard) course page. A proposal template is also available in the same folder on the MyAberdeen course page to help you develop your software engineering project proposal. The proposal submission is mandatory, but it does not carry any marks.

Penalty for Non-Submission or Late Submission: A proposal submitted after the deadline but within 7 days will be subject to the applicable late-submission penalty. A proposal submitted more than 7 days after the deadline will be treated as not submitted. Non-submission or late submission may result in a maximum penalty of 10% of the overall group project mark.

The project proposal should include the following (*please see the proposal template provided on MyAberdeen for further details*)**:**

- Proposal title page: The proposal should include a title page containing the group number, programme of study (BSc (CS), BSc (AI), or BSc (BMIS)), and the student ID and name of each group member.
- Project Title: Give a clear and meaningful title that reflects the main purpose of your proposed software engineering project.
- Problem: Briefly explain the problem your project is intended to address and why it is important.
- Solution: Describe the proposed software solution and explain how it will address the identified problem.
- Objectives: State the main specific and achievable objectives that the project aims to accomplish.
- Benefits: Explain the expected benefits of the proposed solution to its intended users, customers, organisation, or other relevant stakeholders.
- Timeline: Provide an estimated schedule showing the main project phases, activities and expected completion dates.
- Action plan: List the specific activities required to achieve each objective, including group member(s) responsible for each activity, the activity deadline and the current progress status (if any).

Note: *The proposal is expected to be between 4 and 8 pages, and the proposal title page does not count towards this limit.*

Academic Integrity: All submitted work should represent the work of your group. Any material, code, figures, images, data, ideas or other content taken or adapted from external sources must be appropriately acknowledged and referenced. All submissions will be run through Turnitin to check for plagiarism and dealt with accordingly. Be sure to follow guidelines on citing references, and paraphrase what you find elsewhere to avoid your submission being downgraded for plagiarism. Any use of AI tools should follow the University's guidance. The University Academic Integrity page provides all the relevant guidance.

Formatting Requirements: The project proposal and technical report should follow these formatting requirements:

- A4 page size
- Standard margins of 1 inch (2.54 cm) on all four sides (top, bottom, left and right)
- Single-column layout

- 12-point font size
- Arial or Times New Roman font type
- 1.5 line spacing

Submission Format: Follow the formatting requirements specified above and submit your project proposal as a single PDF file on the MyAberdeen course page through the relevant submission area.

Project Suitability and Supervisor Discussion: Although the project proposal is not a marked component**,** you should discuss the proposed project with the assigned academic supervisor to help ensure that its scope, objectives, and planned work are appropriate and feasible for the course group project. Any substantial subsequent change to the agreed project scope or objectives should also be discussed with the academic supervisor.

2. Technical Report (Marked – 50% of the Group Project Mark)

Each group must submit a technical report by the deadline specified in the Course Assessment folder on the MyAberdeen (Blackboard) course page. The technical report is a marked component and contributes 50% of the overall group project mark.

Technical Report Structure: The technical report should begin with a Project Title Page containing the project title, group number, programme of study (BSc (CS), BSc (AI), or BSc (BMIS)) and the student ID and name of each group member, followed by a Table of Contents, List of Figures and List of Tables.

The main content of the report should include the following:

- Introduction and Project Context: Explain the problem, relevant background, project objectives, intended users or stakeholders, main features and potential practical or business value.
- Functional and Non-Functional Requirements: Specify the main system functions and relevant quality requirements, such as usability, performance, reliability, security and maintainability.
- Use Cases and Requirements Modelling: Include appropriate use case diagram(s) and descriptions of the key use cases.
- System Design and Architecture: Describe the system architecture, key components, their relationships and important design decisions. Include appropriate models and diagrams where relevant, such as context models, business process models, architecture diagrams, class diagrams and sequence diagrams.
- Implementation: Explain the Proof-of-Concept implementation, including relevant technologies, tools, frameworks, software structure, algorithms, important technical decisions and graphical user interface design and implementation, where applicable.
- Testing and Evaluation: Include appropriate test cases, testing methods and results and evaluate the software against the stated functional and non-functional requirements. Testing may include unit testing, component testing, interface testing and performance testing, where relevant.
- Conclusions and Future Work: Summarise the main project outcomes, limitations, and realistic opportunities for further development.
- References: Provide a complete list of all cited sources using a consistent academic referencing style.
- Appendix A – Group Workload Profile:
    - Complete the group workload profile using the template provided on MyAberdeen and include it as Appendix A. The purpose of the Group Workload Profile is to ensure fairness in the grade individualisation process, starting from the grade awarded to each group project submission. Members of each group must record their effort by openly and fairly agreeing on the quantitative and qualitative contributions of each group member during the project.
    - A group member may not agree with one or more contribution values agreed by other members in the profile. In such cases, they should:
        - Avoid signing the profile, as signing implies unconditional acceptance of ALL that is written in it concerning each group member.
        - By the project submission deadline, email the Course Coordinator explaining the reason for refusing to sign the profile.

Where there is no disagreement, the group must ensure that all team members have signed the form and agree with the declaration before submitting it with the coursework.

Group members may complete the profile using the following values to indicate their relative effort:

- 0% - I contributed the same as my team members.
- +X% - I contributed more than my team members by a specified amount X%.
- -X% - I contributed less than my team members by a specified amount X%.

For example, X% = 10%, 25%, 50% or 100%.

Note: *The technical report is expected to have a minimum of 40 pages and a maximum of 60 pages. The page count applies to the main content of the report, from the "Introduction and Project Context" to the "Conclusions and Future Work" sections. The Project Title Page, Table of Contents, List of Figures, List of Tables, References and Appendix A – Group Workload Profile do not count towards this page count.*

Source Code: Do not include source code in the technical report; instead, describe and explain only the relevant implementation details, algorithms, and software structure.

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- Single-column layout
- 12-point font size
- Arial or Times New Roman font type
- 1.5 line spacing

Submission Format: Follow the formatting requirements specified above and submit your technical report as a single PDF file on the MyAberdeen course page through the relevant submission area.

What the Examiner Will be Looking for:

- Problem and Project Context: Clear explanation of the problem, relevant background, objectives, intended users, main features and potential value.
- Requirements and Use Cases: Appropriate functional and non-functional requirements, supported by relevant use cases and requirements modelling.
- System Design and Architecture: Clear system design and architecture, with suitable models, diagrams and justified technical decisions.

- Implementation: Clear explanation of the Proof-of-Concept implementation, including relevant technologies, software structure, algorithms, technical choices and user interface (where applicable).
- Testing and Evaluation: Appropriate unit, component, interface, performance, or other relevant testing, with suitable test cases and results demonstrating how well the software satisfies the stated requirements.
- Consistency and Technical Depth: Clear alignment between the problem, requirements, modelling, design, implementation and testing.
- Conclusions and Future Work: Appropriate reflection on project outcomes, limitations and realistic future development.
- Writing, Presentation and Referencing: Clear, well-structured presented work with appropriate figures, tables, and consistent academic referencing.

3. Proof-of-Concept Software (Marked – 30% of the Group Project Mark)

Each group must submit Proof-of-Concept (PoC) Software by the deadline specified in the Course Assessment folder on the MyAberdeen (Blackboard) course page. The Proof-of-Concept Software is a marked component and contributes 30% of the overall group project mark.

The PoC should demonstrate that the proposed software solution is viable and works in practice. It should implement the core features, algorithms, and components required to demonstrate the project's main objectives. The PoC does not need to be a complete production-ready system, and its scope should be appropriate to the nature of the project.

The Proof-of-Concept submission should include the following:

- Program Code: Provide the complete source code required to run and evaluate the Proof-of-Concept Software. This should demonstrate the core features and functionality required by the main project objectives.
- Executable/Runnable Software: Provide an executable or other suitable runnable version of the software, together with any files, libraries, data or dependencies required for execution.
- User Manual: Include a user manual containing clear instructions on how to install, configure, run and use the software. The user manual should be 8-12 pages and submitted as a PDF file.

Scope of the Proof-of-Concept: The PoC may use simplified or prototype components where appropriate. For example, you may use a fixed dataset instead of a full database or provide a simplified interface when the user interface is not central to the project. Discuss and agree on the exact scope with the assigned academic supervisor.

Academic Integrity: All submitted work should represent the work of your group. Any material, code, figures, images, data, ideas or other content taken or adapted from external sources must be appropriately acknowledged and referenced. Be sure to follow guidelines on citing references, and paraphrase what you find elsewhere to avoid your submission being downgraded for plagiarism. Any use of AI tools should follow the University's guidance. The University Academic Integrity page provides all the relevant guidance.

Submission Format: Submit one ZIP file containing:

- Program source code
- Executable or other runnable version of the software
- Any required supporting files or dependencies
- User manual in PDF format.

What the Examiner Will be Looking for:

- Functional Completeness: Successful implementation of the core features and functionality required by the project objectives.
- Code Quality and Standards: Readable, modular and well-structured code that follows appropriate coding standards and good software engineering practices.
- User Interface and Experience: An appropriate, usable and intuitive interface, where applicable.
- Reliability and Stability: Software that runs reliably without critical bugs or crashes and demonstrates appropriate stability.

- Consistency with Technical Report: Clear alignment between the stated requirements, design, implementation and testing in the technical report and in the submitted Proof-of-Concept Software.
- Documentation and Reproducibility: Clear instructions and sufficient supporting material to allow the examiner to install, run and use the software.

4. Project Presentation (Marked – 20% of the Group Project Mark)

Each group must submit a Project Presentation by the deadline specified in the Course Assessment folder on the MyAberdeen (Blackboard) course page. The Project Presentation is a marked component and contributes 20% of the overall group project mark.

The presentation should provide a concise, well-structured overview of the software engineering project, explaining the project objectives, development approach, key outcomes, and the main features of the Proof-of-Concept Software. Each group will have 15 minutes to deliver the presentation.

The Project Presentation should include the following:

- Introduction: Provide a brief overview of the project, including the problem, objectives, and proposed solution.
- Project Approach and Key Outcomes: Summarise the main development approach, important technical or design decisions and key project outcomes.
- Main Demonstration: Provide a demonstration of the Proof-of-Concept Software, highlighting its key features and functions.
- Conclusions and Future Work: Summarise the main conclusions, limitations and realistic opportunities for further development.

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Submission Format: Submit one ZIP file containing:

- PowerPoint presentation slides.
- A recording of the presentation in MP4 format. The recording must include a presenter(s) appearing while delivering the presentation.

What the Examiner Will be Looking for:

- Content Coverage: Clear and appropriate coverage of the project objectives, development approach, key outcomes, Proof-of-Concept demonstration and conclusions.
- Demonstration: A clear and effective demonstration of the core features and functionality of the Proof-of-Concept Software.
- Delivery and Communication: Clear, confident and effective communication of both technical and non-technical information.
- Visual Aids and Organisation: Well-structured and professionally presented slides and visual aids that support understanding.
- Team Participation: Participation and contribution from all group members.
- Consistency with the Project: Clear alignment between presentation, technical report and submitted proof-of-concept software.

5. Project Updates (Mandatory – Unmarked)

The group and the assigned academic supervisor use project updates to monitor progress and facilitate discussion. One group member is required to submit the project update on behalf of the group. Four project updates will be scheduled during the project, and each group must submit at least three. To support the project update submissions, online forms have been created on the MyAberdeen course page under the assessment section for the group project.

Each project update should include the following:

- Summary of the Work Done to Date: Briefly describe the work completed and the progress made since the previous update.
- Challenges faced and solutions: Identify any important challenges or problems encountered and explain how these have been addressed or how the group plans to address them.
- Next Steps: Briefly describe the main activities that the group plans to complete next.

The project update should reflect the group's actual progress and the discussion with the assigned academic supervisor during project development.

Penalty for Non-Submission or Late Submission: A required update submitted after its deadline but within 7 days will be subject to the applicable late-submission penalty. An update submitted more than 7 days after its deadline will be treated as not submitted. Failure to submit at least three project updates may result in a maximum penalty of 10% of the overall group project mark.